

Governing AI-Driven Drug Discovery: A Policy Framework for the Ethical Use of AlphaFold3 in Drug Candidate Selection

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Problem Identification & Ethical Context:

3-D models using neural network based on deep learning algorithm of protein structures provide faster method of selecting drug candidates, compared to traditional method of looking at amino acid sequence to determine protein structures to identify binding sites for drugs.

AlphaFold2 has predicted 200 million protein structures, a neural network model, while AlphaFold3 is the newer model that expands this work by predicting biomolecular structures and interactions relevant to drug discovery. To support this pioneering work, an extensive database of protein structure is the AlphaFold Protein Structure Database.¹ The R&D process is being rewritten for the pharmaceutical industry with this algorithm. Drug development is time consuming and costly endeavor for any company to build its pipeline for a given disease or a target therapeutic area.

Producing a compound comes with an astronomical price tag. To produce one drug that is considered safe and efficacious cost over 2 billion dollars to achieve regulatory approval from health authority(ies).² It takes about 10-15 years for a full lifecycle for a drug to be developed for marketing.³ Given the high cost of drug development and lengthy lifecycle for R&D process for a compound, the need for more efficient means of developing a drug is noted to reduce the cost and development time, using AI.⁴ This is where AlphaFold3 comes in with its ability to screen for drug candidates faster and more efficiently than just looking at amino acid sequence to predict 3-D structure of proteins. Thus, this algorithm provides faster means of drug candidate

identification by understanding predicted protein structures' potential protein-ligand interaction sites faster. The target drug in development would be the ligand.⁵

Given the benefits of AlphaFold3, considerations of its implementation would need to be beyond just the technical aspects of this model. Ethical considerations need to be addressed as well. Notably, there is no verification that any drug that would be approved for human use developed using AlphaFold3 would be as safe and effective as those developed using a traditional method of protein-drug interaction models. There are no drugs that have been approved for human use using this algorithm yet. This is due to the lengthy timeline for R&D process for any drug to get market approval by health authorities. This means there is no data available on how safe AlphaFold3 drug candidate selection process is for a given compound for development that is ultimately for the public consumption. This is a risk that the pharmaceutical industry is willing to take given the benefits of this technology, as such several drug companies are partnering with Isomorphic Labs (Novartis and Eli Lilly) to develop small molecules using AlphaFold3 for drug selection,⁶ but this comes with a cost to the public with uncertainty of safety and efficacy concerns of compounds selected for protein targets with this AI. This is evident by the fact that the AI algorithm cannot predict mutations or post-translational modifications of proteins. It also does not have full grasp of interactions in a biological system what molecules proteins interact with.⁷ Thus, this lack of clarity on drugs made using AlphaFold3 brings forth serious concerns of safety and efficacy of compounds.

Additionally, at some point, drugs approved would be developed using AlphaFold3 technology, but currently, there is no regulatory framework to suggest that patients would be informed if their prescriptions were made using an AI algorithm. They would not have the option of choosing a drug developed using a traditional method, or using an AI algorithm, if both

options do exist. This raises a concern of does the public have the right to know of the possible risks they are being exposed to from a novel technology use for drug development. Another ethical issue is that in order for a drug to be approved for marketing, it needs to go through clinical trials. At this point, there is no clear indication that the subjects would be informed that they would be taking drugs made using an AI model, with unknown safety profile given the lack of any history on such matter since there is no real-world data on this at all. Even participants in research would be affected by this uncertainty of safety. However, there is no requirement that they would be informed of test drugs they are taking is made from an AI model.

The informed consent process for this is just not there yet to protect subjects in clinical trials for drugs made using AI technology. On the other hand, companies do have the right to protect their trade secrets of not disclosing that the test drugs/potential drugs for the market have been developed using AlphaFold3. This may be needed to maintain their competitive edge, so competitors do not copy companies using AlphaFold3 technology for drug candidate selection that targets the same disease being studied in research or would be marked for human use. The question then becomes what is more important in these types of scenarios, the public's awareness of AI use in drug candidate selection for possible safety concerns or the pharmaceutical industry's need to protect their interest.

As such, these ethical concerns and the question of whether the public's safety or the pharmaceutical industry's interest have greater importance lends themselves to ethical principles use to provide a guide for possible next steps for these issues. For drugs that potentially could have safety and efficacy issue selected using AlphaFold3, consequence-based theory would apply, utilitarianism, with uncertain safety and efficacy profile is morally not justifiable. It can harm individuals and create risks, just to further the bottom-line of the pharmaceutical industry

for a need of cheaper drug development process. This is not in-line with utilitarianism of producing greatest good per Tavani,⁸ so a layer of safety protection would need to be added for drugs made using AlphaFold3.

For other ethical concerns, different ethical frameworks for them. Not informing the public of potentially unsafe drugs in the market, which can lead consumers to take medications with unknown risks falls under contract-based theory, namely social contract theory. Individuals have the right to know this information. In clinical trials, participants need to be informed of the risks of the study they are participating in through the informed consent process. This is required by regulations by the health authorities; for the FDA, it is codified in 21CFR.⁹ Based on duty-based theory, humans should not be treated as means to an end.⁸ The lack of public disclosure and improper consent for research participants puts individuals not just at safety risk, but their autonomy to make informed decision is not provided. For the last item, pharmaceutical companies protecting their trade secrets vs. public welfare, character-based theory applies, namely virtue ethics. It is moral to protect the public from safety risks, by providing information to consumers of potential risk of drugs made using AI technology rather than protecting trade secrets.

Stakeholder Analysis:

Given these various aspects for consideration for AlphaFold3 use, it is not surprising that there is a long list for all relevant stakeholders. The list includes pharmaceutical industry, health authorities, ICH, patients/consumers, healthcare professionals, private insurance companies, and public insurance for federal and state governments (Medicare/Medicaid). For each stakeholder, their interests, needs, and concerns would need to be understood to gain knowledge of what it means for them this new technology is now taking part in drug development space.

The stakeholder that drives the drug development itself is the pharmaceutical industry. It has become increasingly costly to produce a drug to achieve market approval. In 2003, it cost \$802 million to produce a compound for its full lifecycle of R&D. This amount grew to \$2.3 billion in 2023.¹⁰ Pharmaceutical industry's utilization of AlphaFold3 would reduce R&D price for drug development by predicting drug target sites for proteins in seconds instead of years by traditional means.⁵ This would speed the development of drugs, with faster protein-ligand identification. Still, this does come with concerns for the drug industry. Drug companies have their own proprietary data that they hold for protein structures; however, these companies are not sharing this information with their competing drug companies.¹¹ This industry is concerned about competition in the marketplace, which is resulting in a barrier to progress of 3-D modeling of protein structures.

Health authorities are responsible for safeguarding the public by providing consumers with safe and effective drugs. The gold standard for a health authority is the US FDA. As such, the FDA has initiated first steps in its current thinking in AI regulation, in the form of guidance document that is in draft stage currently.¹² This indicates that the Agency started thinking about what AI use means for drug development and public safety. This is an indication that it recognizes the need to regulate AI in the drug industry. So far, nothing has been codified for this topic in 21CFR, yet. Also, this current thinking by the FDA doesn't discuss how AlphaFold3 should be used in drug development, for technical or ethical considerations. Without an existing regulatory framework for the use of this algorithm, an agreement among health authorities does not exist on how to manage AlphaFold3 from regulatory perspective. This is something that would need to happen given that drugs are generally marketed globally. The FDA would need to take this on first as the gold standard for a health authority for other health authorities to follow.

The concern is that other countries will limit the use of AI models, given some countries are moving forward with regulating AI, while US is not. This could potentially limit the use of AlphaFold3 globally.

Since regulations that apply globally for this technology do not exist, ICH guidelines can fill in the gap for this: ICH applies globally across the drug industry. This committee can convene to discuss the global interest to fully implement the use of this technology. This would serve the interest of ICH well to address the concerns that are ethical in nature by filling in regulatory gaps with its industry guidelines. The ICH can then fulfill its existence to harmonize drug development process, in this case the use of AlphaFold3.

By looking at these three stakeholders first, it would help to look at the remaining list of stakeholder analysis. Patient/consumer and healthcare professional relevance to this new technology are closely tied together. Healthcare professionals would need the knowledge to prescribe drugs once developed using this algorithm and provide proper guidance to patients on safety/risk profile for them. Having a proper regulatory framework would help support this early-on when these drugs do become approved for market use eventually. This would allow patients to make informed decisions on if such medications would be appropriate for them with the guidance from their doctors. The only problem with this is that if the patient becomes worried about drugs made with new technology and is too apprehensive about taking these drugs when no other option is available in the market, it can lead to non-compliance of medication due to fear of risk that may be unsubstantiated. Such consequences can also mean healthcare professionals would need to better understand patients' concerns to support medication compliance. To mitigate this potential unintended consequence, some form of public education

on safety of AlphaFold3 may need to be done to ease the minds of patients/consumers and support doctors' agenda to promote the patients' health.

As a final note, reduced drug development cost could also impact insurance, private and public. The need to manage health insurance costs is evident. "In 2023, overall pharmaceutical expenditures in the U.S. grew by 13.6% compared to the previous year, reaching a total of \$722.5 billion."¹³ As such, cost of drugs is one of the major drivers of rising cost in insurance for Medicaid, Medicare, and commercial insurance. The impact of AlphaFold3 by lowering R&D costs for drug development should help with this issue. The local and federal government, in addition to private insurance companies, have interest in seeing this to realization. Only issue is that there is no guarantee that the cost-saving measures possibly seen by this new technology would be passed on as lower cost of drugs for insurance companies. There is the possibility that these savings would go back into the drug industry's profits. If that happens, consumers will not see the benefit of AlphaFold3 for their healthcare insurance premiums or co-pays for drugs. For Medicaid, the state and federal government price tag may not change as well.

Policy Advocacy & Feasibility:

Given that concerns for AlphaFold3 utilization is global, there needs to be an international regulatory perspective in place. The only option there is to place something at that scale would be through ICH guidelines. ICH can provide drug development guidance that spans the globe, given its authority to do such, as it pertains to AlphaFold3 concerns from its utilization. These guidelines would not be regulations; however, they would carry the weight of regulations, so the pharmaceutical industry would need to adhere to them, given ICH's overarching authority over GxP push in the pharmaceutical industry, with its guidelines. Additionally, health authorities can adopt these guidelines as regulations when permitted.

An example of ICH guidelines that are used by the drug industry is for GCP, Good Clinical Practice, that dictates how clinical trials should be conducted with human test subjects. This guideline is adopted internationally for conducting clinical trials.¹⁴ This is a living document that gets updated periodically. Given the success of ICH's role in clinical research and its ability to harmonize regulatory framework through guidelines, AlphaFold3 can be built with regulatory framework through guidelines initially as well. In fact, approaching ICH as the first step is necessary given that each country's health authority functions slightly differently with various regulatory requirements.

As a first step, ICH should meet as a committee to discuss the agenda that needs harmonizing for AlphaFold3 concerns, which are ethical as well as the need to incorporate the various stakeholders' agendas. This would serve as the policy needed at this scale. So, in this context, the upcoming guideline and new policy can be synonymous. ICH should call this document: "Guideline: AI algorithm adoption to support drug candidate selection with predictive modeling of protein structures to reduce R&D timeline, with ethical considerations that has multi-layer approach to meet the needs of stakeholders."

The pharmaceutical industry can take the policy to address ethical issues. Currently, human subjects are not told during the informed consent process how a drug is made. This is something that would need to be added part of the discussion for informed consent, which they would sign to be in a clinical trial.¹⁵ This is imperative for drugs made using AlphaFold3, due to uncertainty of level of risk for drugs selected using this algorithm. Some type of Black Box Warning¹⁶ should be on drugs once marketed with disclaimer that AlphaFold3 was used to select the drug candidate, with additional risk levels being observed in real world. The public safety should be paramount,¹⁷ and it should be made clear in this document that it overrides trademark

secret risks, when these two competing items exist. By following these policy requirements, public safety and autonomy would be adhered to resulting in resolution of ethical concerns discussed previously. This policy should also include a comprehensive framework for GxP impact in general as well for the drug industry with this technology implementation, to ensure it fills in any gaps needed for compounds made using this algorithm from regulatory perspective.

The FDA can then take this policy and build guidance documents that get pushed to the drug industry, until regulations catch up. The Agency can use guidance documents and this policy to aid in drug development efforts that are properly aligned from regulatory perspective to support AlphaFold3 involvement. Other health authorities can borrow from these lessons that the FDA would be following, to support their need for efficient means of drug research. Once all this is in place, patients/consumers and healthcare professionals would be armed with sufficient information to better understand the risk of these types of drugs to help support prescribing needs as well as medication compliance by patients. AlphaFold3 produced drugs would mean lower cost of drug production eventually when marketed. To ensure that this is passed down to consumers in savings for healthcare insurance, the federal government would need to pass a bill to guarantee this. Some type of patent limit law should also be put in place for companies to share data on protein structures they have as well to remove barrier for this innovative work progress.

Critique & Unintended Consequences:

The pharmaceutical industry needs drugs that make it market to carry those that fail. They hope: “the drug that gains market approval” is a blockbuster, a billion-dollar drug. This expectation is necessary for a drug company’s survival, given 90 percent of drugs that undergo development fail.¹⁸ With such statistics, if drug prices are controlled through legislative means to

regulate health insurance cost, the drug industry may face issues of sustainability as an industry. High profit of successful drugs maintains this business. One might argue that AlphaFold3 would reduce the cost of R&D to make up for this. There isn't any data to verify this, or how much cost control would be beneficial to the insurance industry, while not jeopardizing the drug companies' bottom-line. Before enacting any legislative act to control drug prices, some data is to understand the right price point of these drugs made using this new technology. Price control of these drugs can be done to manage it properly for the drug industry to benefit the healthcare insurance sector.

Another consequence of the policy is related to an update to informed consent process for clinical trial participants. Human subjects have the right to know the risks involved with test drug, for AlphaFold3 derived drugs in particular. Who will be qualified to answer all questions raised by these subjects. How will the education on these drugs will be pushed to clinical research sites of the unknown risk from this new technology use. There is the possibility that this medical education would be insufficient, and subjects may become fearful of taking drugs made using this algorithm. It could lead to recruitment and retention challenges in studies. This is a known industry-wide problem, but it could be made worse by this situation. Thus, the gains made by speedy drug-candidate selections could be lost during the clinical trial phase due to subjects being hesitant on taking such compounds. It could also erode the gains that would have been achieved in cost reduction as well. To mitigate such potential obstacles in research, recruitment and retention should be tailored to provide extensive education on the balance of risk/benefit ratio for these compounds. "Policy How" can further now give credibility to this paper, as a conclusion.

Policy How (part of Policy Advocacy & Feasibility):

The ICH guideline for AlphaFold3 use should relate to ethical concerns and stakeholder relevance (policy) would basically be an update to GxPs being used in the pharmaceutical industry, mainly for GCP and GMP.

Clinical trials follow GCP for subjects enrolled in research. Once the potential participants are contacted through recruitment efforts, the informed consent process begins. This is when the policy efforts should start. The ICH guideline would need to indicate that initiation of the informed consent process needs to provide potential patients that the study would involve investigational products created using an AI model. The sites would be required to provide information on this with what is known in terms of benefit/risk ratio for the study compound. The policy should have a script that is generic to support this type of contact for potential patients. In addition, the policy should also provide list of Q&A that could aid the sites with this type of communication, so that study coordinator/doctors at the investigational sites can guide the patients with proper understanding of what this technology use could mean for the drugs in use. Additionally, the policy should have additional wording needed for the actual informed consent form template that would be added to site-level ICF forms that has the details of this model use in research drugs, that would ultimately be signed by patients.

Once these drugs are approved for marketing, GMP for labeling should kick-in. The standards for the Black Box Warnings for these drugs would need to be included in this policy. This would include the wording of the warning, and dimensions and the color of the box. Additionally, the policy should require packing inserts to have details on who to contact for more information related to the warning for patients and healthcare professionals and a requirement for a link to a website that provides details on what AlphaFold3 use means for drug development as GMP portion of the policy.

With this “Policy how” section, ethical concerns are addressed for safety, public welfare, healthcare professional/patients would be able to make informed decision on the use of drugs made using AI, trade secrets do not overtake public safety, patients in research have autonomy to decide if research drugs using AI are appropriate for them, and health authorities’ agenda of safeguarding the public is adhered to for drugs made using AlphaFold3.

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Appendix:

1. Therapeutic area: Terminology used in the pharmaceutical industry to really mean particular area of medicine, such as hematology, ophthalmology, etc.
2. Health authority(ies): it could mean the FDA or another country version of the FDA

3. Regulatory approval: this is when health authority(ies) approve drugs for human use.
4. FDA: US FDA (Food and Drug Administration: US health authority)
5. R&D: Research & Development
6. Research phase in humans: this refers to clinical trials where test subjects are humans.
7. 21CFR: FDA regulations (CFR: Code of Federal Regulations)
8. ICH: it is a committee, and not a regulatory body/health authority. They provide guidelines to the pharmaceutical industry for the conduct of clinical trials.
9. ICH: The International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (Harmonisation: this is correct spelling)
10. FDA's guidance documents carry the weight of regulations, and it would need to be adhered to even in draft form. They represent FDA's current thinking on the topics represented in those documents.
11. Agency: refers to the FDA.
12. GxPs: required regulations and guidelines that pharmaceutical industry must follow during the lifecycle of a drug during development and while in the market for consumer use as well.
13. Study: is basically a clinical trial.
14. Note: AlphaFold3 is the newest algorithm. Previously, it was AlphaFold2.
15. GMP: Good Manufacturing Practice (This is the standard used in production for drugs intended for humans, that ensures quality and safety for human consumption. This is required for drugs.)
16. GCP: Good Clinical Practice (standard for clinical trials to ensure that they adhere to regulations and with safety in mind.)
17. ICF: informed consent form

- 18. Site-level ICF: ICF for the location where the clinical trial is being conducted
- 19. Study drug/investigational drug: drug being researched during the clinical trial
- 20. Site: location where clinical trial is conducted.
- 21. Subjects/participants: those enrolled in clinical trials.
- 22. Small molecules: drugs